

Behind birds eyes: An exploration of the relationship between eye orbit morphology, phylogeny and feeding in passerides

Sophie Shindler

INTRODUCTION

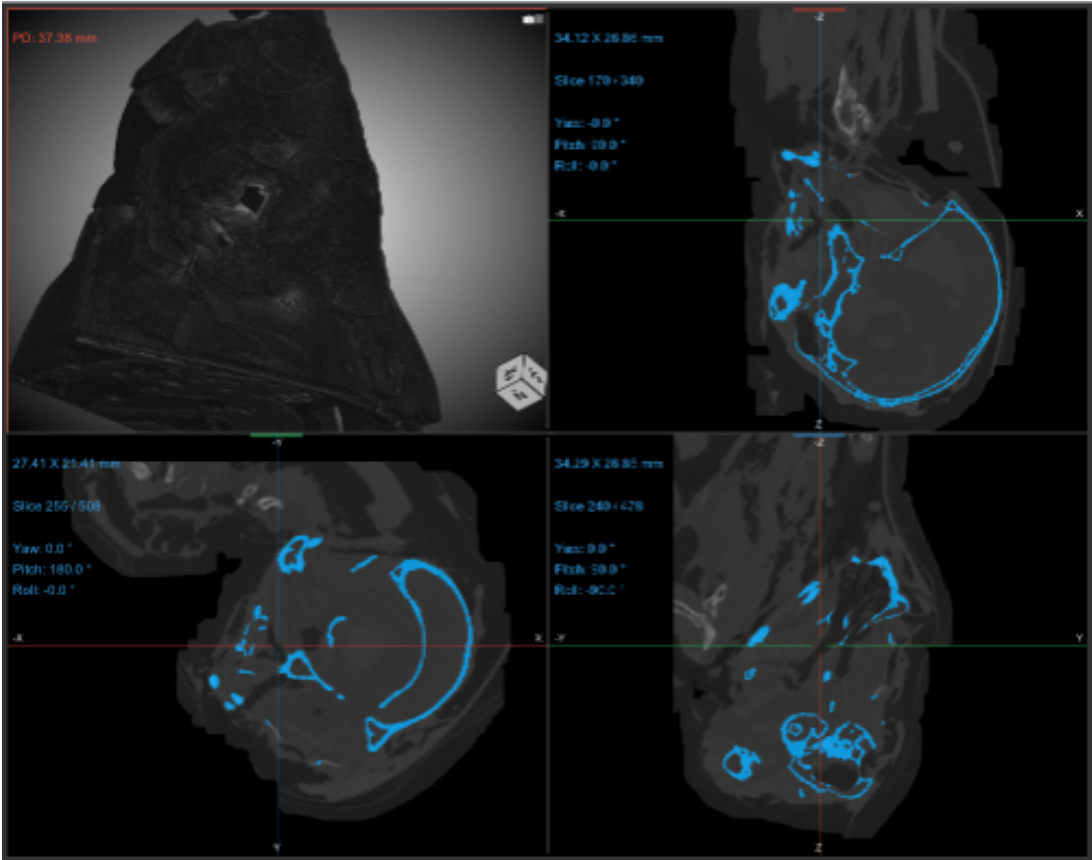
- Birds rely heavily on vision, particularly passerines¹
- Despite this, variation in eye orbit morphology is relatively understudied²
- Recent research suggests that cranium shape is influenced more by phylogeny than ecology (diet)⁴
- It is unclear whether eye orbit morphology variation is better explained by evolutionary history or ecological function in passerides**



OBJECTIVES

- This project aims to test the hypothesis that phylogeny is the primary driver of eye orbit morphology
- Variation will be quantified and compared across ecological and phylogenetic groups, looking for significance
- This variation can then be related to vision in a biological context within passerides

METHODS

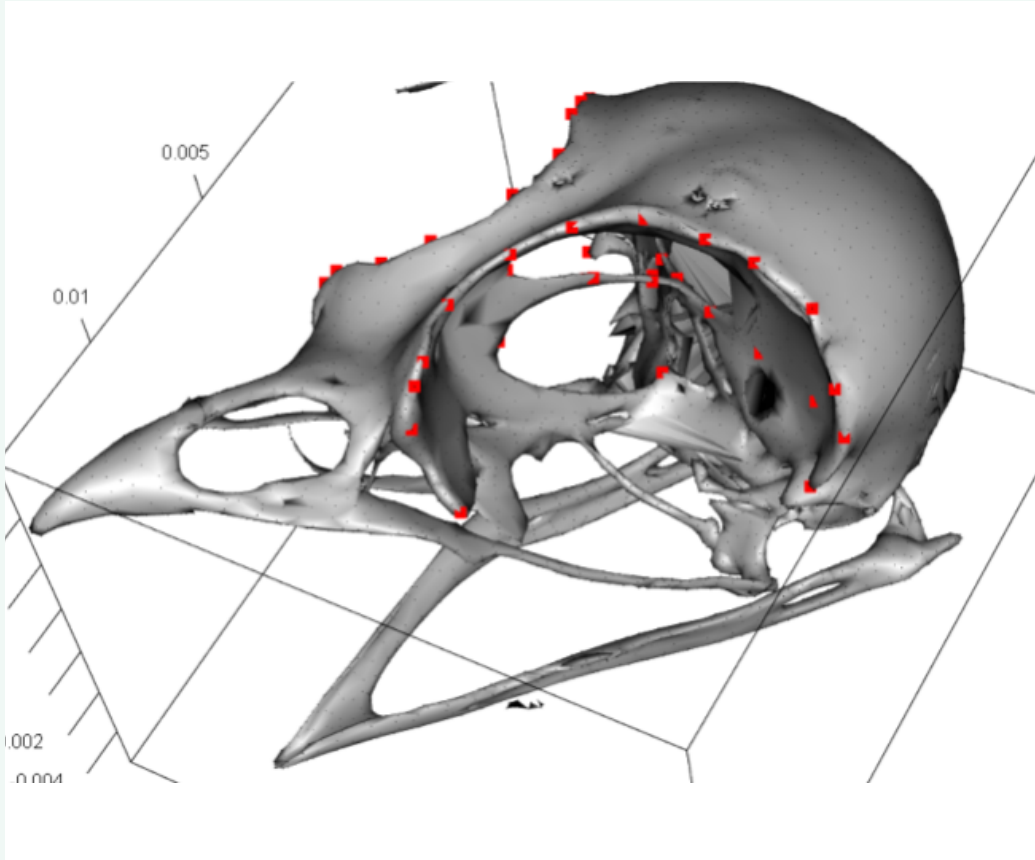


Obtaining the specimens

- The study involves 15 specimens from across the passerides
- They were downloaded from MorphoSource as whole body or skeleton CT image series or as skull meshes
- CT image series required segmenting in Dragonfly
- The resulting meshes were processed in MeshLab

Landmarking

- 42 landmarks were placed on each skull - 12 fixed landmarks and 30 semi-landmarks
- Landmarks were placed in R using the geomorph package and analysed using the Morpho package



RESULTS

Plots showing the results of Principle Component Analysis (PCA) based on the results of a Generalised Procrustes Alignment (GPA). Plots colour coded by phylogeny

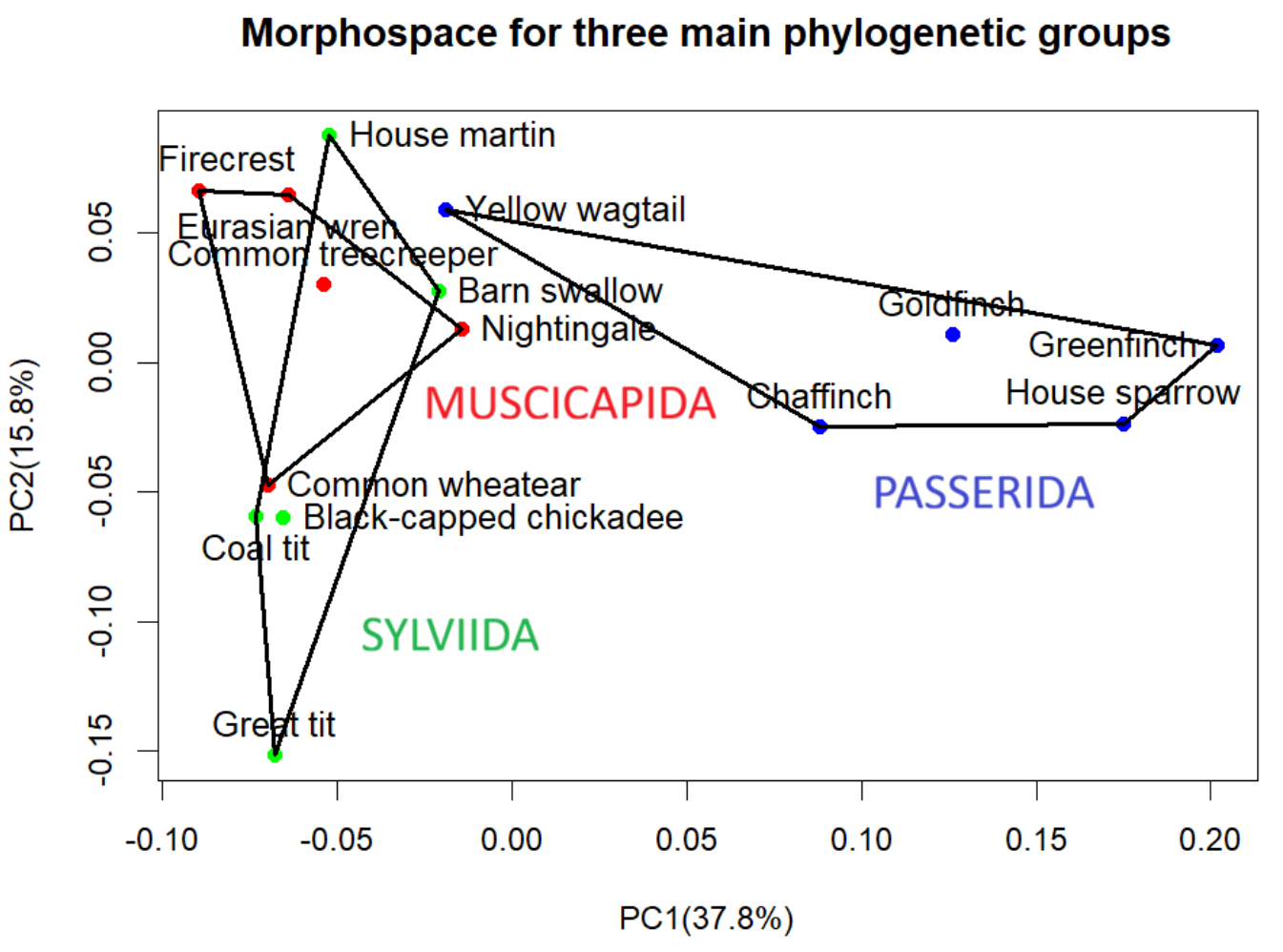


Fig 1: PC1/2, shows separation of Passerida from the other groups, which have overlapping morphospace

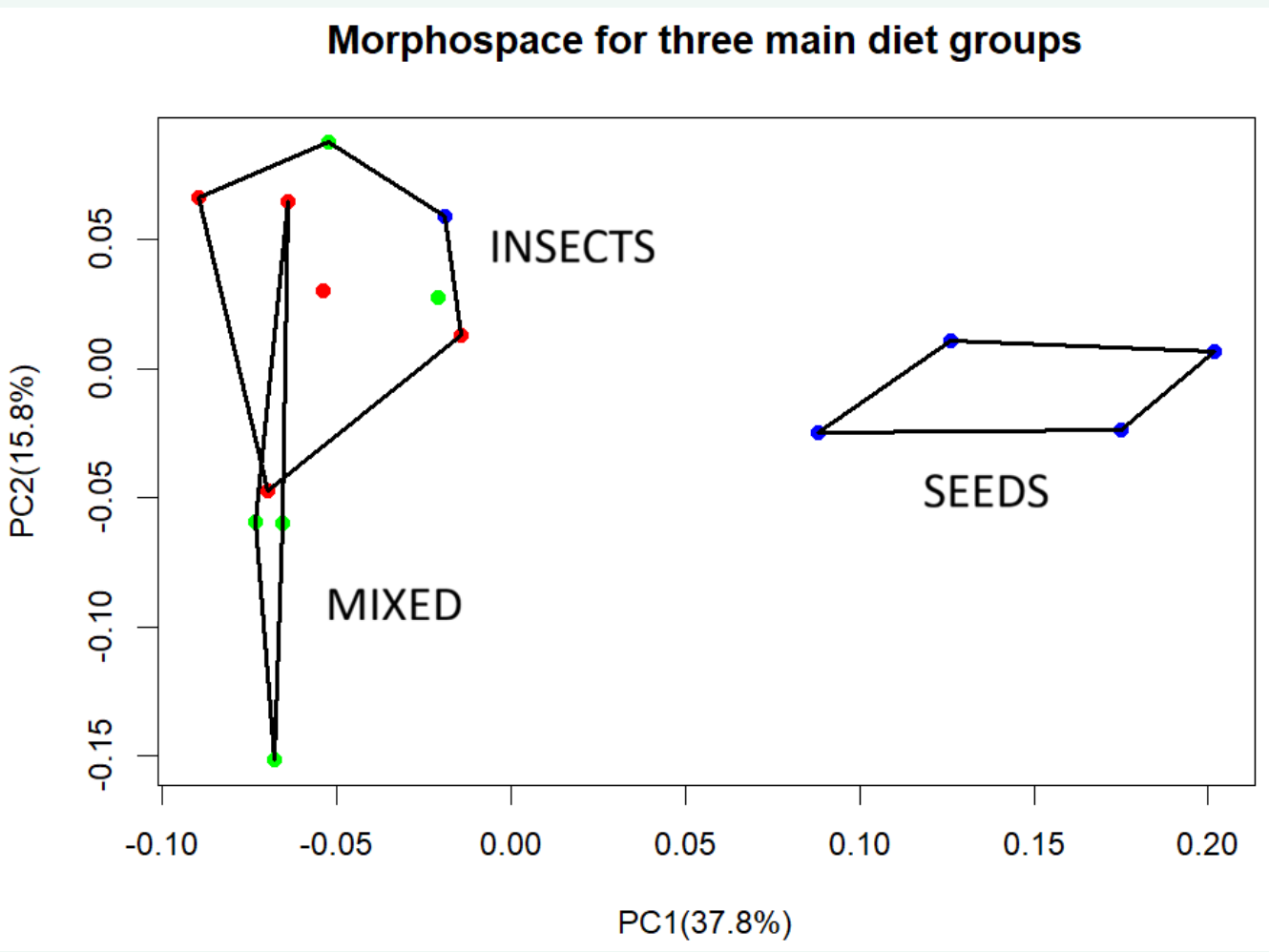


Fig 2: PC1/2, clear separation of seed eaters; insect and mixed feeders have overlapping morphospace

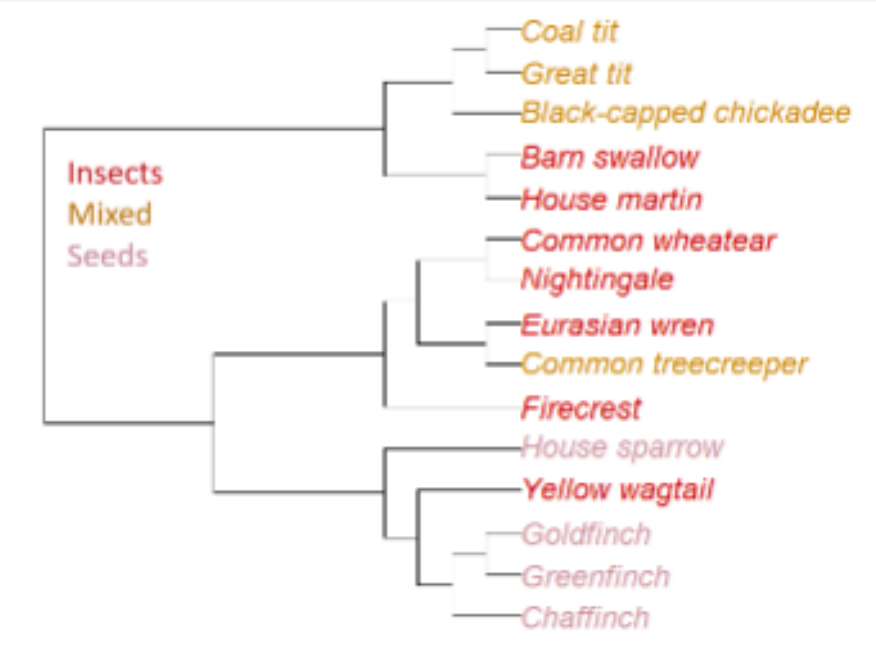


Fig 3: true phylogenetic tree⁵ colour coded by diet

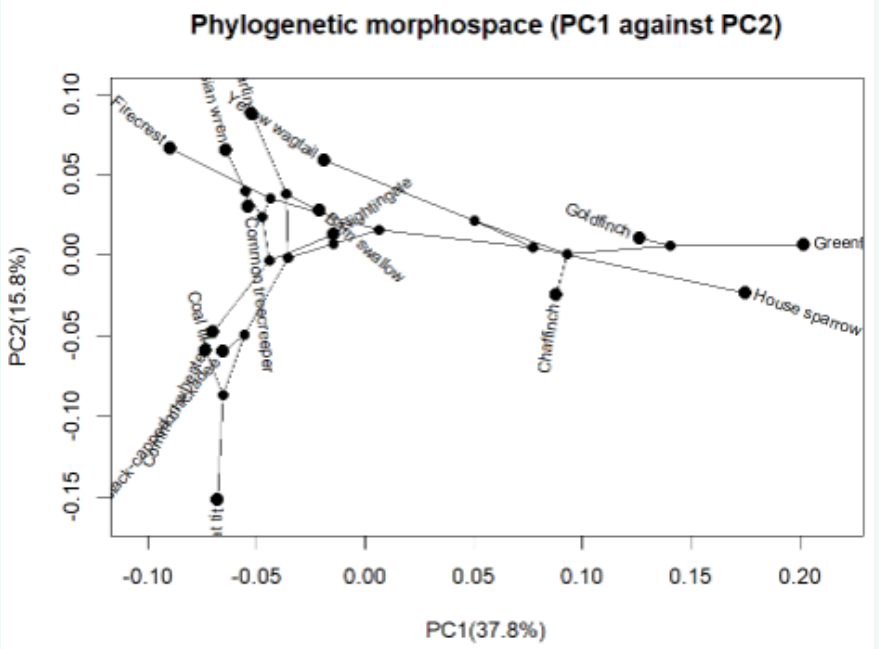


Fig 4: PC1/2, phylomorphospace shows limited clustering

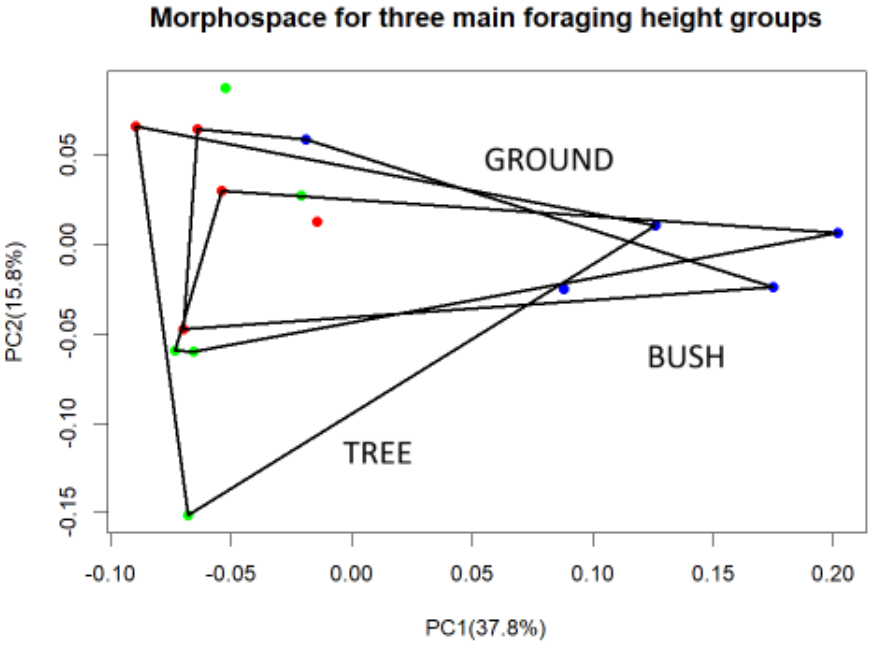


Fig 5: PC1/2, very little separation between any groups

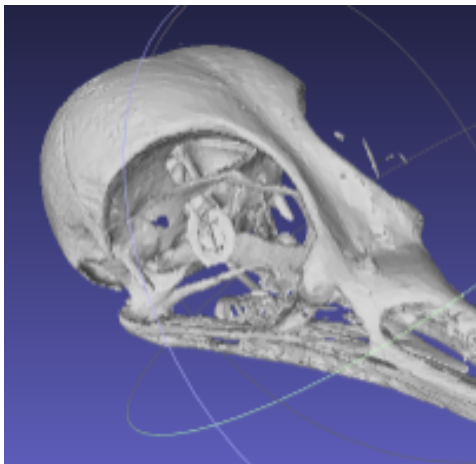
DISCUSSION

PCA results

- PC1 interpreted as an increase in orbit size and decrease in distance between orbits
- PC2 as an increase in interocular hole size
- Insectivores have smaller interocular gaps suggesting eyes are proportionally larger
- This may increase visual acuity¹

Significance tests

- The effect of diet on orbit morphology was significant only when phylogeny was unaccounted for
- Diet explained 20% of the variation in the PGLS despite not being significant
- Foraging height was not significant nor was it affected by phylogeny
- There is an association between diet and phylogeny so it is unclear which is the driver



Limitations

- Small sample size
- Uncertainty in landmark homogeneity
- Exclusion of other skull features
- Failure to separate between different visual demands

CONCLUSION

Diet was associated with eye orbit morphology, but this effect was not significant after accounting for phylogeny. This indicates phylogeny may be a stronger driver of orbit morphology than diet.

ACKNOWLEDGEMENTS/REFERENCES

MorphoSource.org - special thanks to oumnh and fmnh RSPB.org.uk
1. Jones et al, 2007, "Avian Vision: A Review of Form and Function with Special Consideration to Birds of Prey", Ex. Pet Med., 16(2):69-87
2. Sun et al, 2018, "Geometric morphometric analysis of skull shape in the Accipitridae", Zoomorphology, 137:445-456
3. Kulemeyer et al, 2009, "Functional morphology and integration of corvid skull", Front Zool 6(2)
4. Felice et al, 2019 "Dietary niche and the evolution of cranial morphology in birds.", Proc. R. Soc. B, 286
5. Schmitt and Edwards, 2022, "Passerine birds", Current Biology, 32(20):1149-1154

